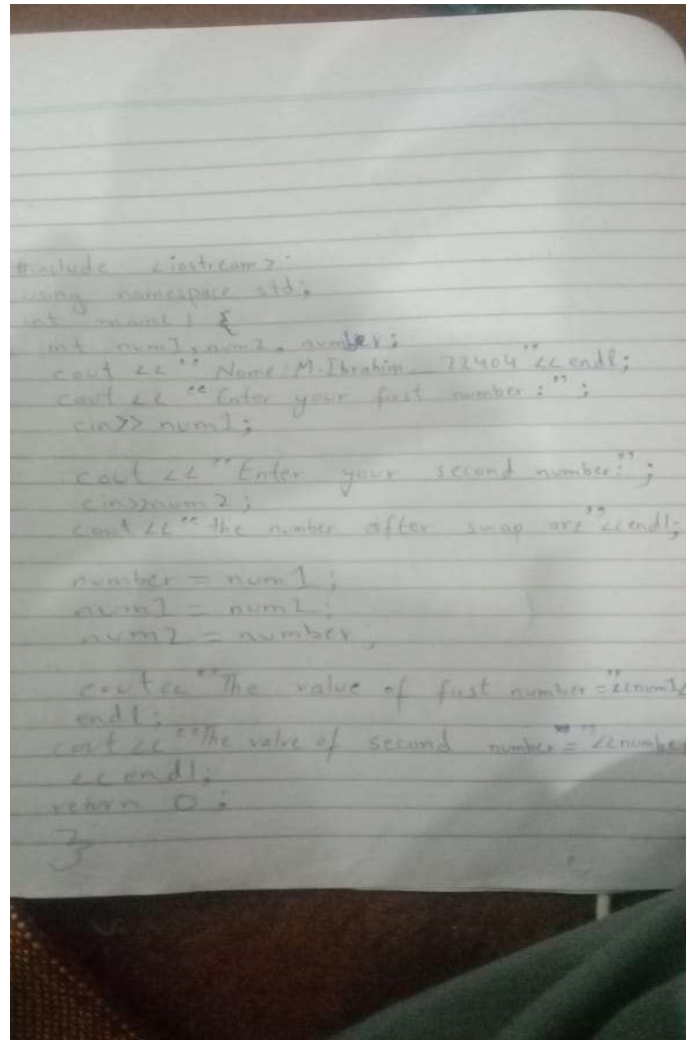


Q 1: Write a program input two numbers, swap the values and then display them.

```
#include <iostream>
using namespace std;
int main() {
    int num1,num2,number;
    cout<<"Name:M.Ibrahim_72404"<<endl;
    cout<<"Enter your first number:";
    cin>>num1;

    cout<<"Enter your second number:";
    cin>>num2;
    cout<<"the number after swap are"<<endl;
    number=num1;
    num1=num2;
    num2=number;
    cout<<"The value of first number = "<<num1<<endl;
    cout<<"The value of second number = "<<number<<endl;
    return 0;
}
```

```
Name:M.Ibrahim_72404
Enter your first number:5
Enter your second number:3
the number after swap are
The value of first number = 3
The value of second number = 5
```



A photograph of a piece of lined paper with handwritten C++ code. The code is a program to swap two numbers. It includes the necessary headers and namespace, declares three integer variables, prompts the user for two numbers, reads them, performs the swap using a temporary variable, and then prints the values before and after the swap. The handwriting is in blue ink and is somewhat cursive.

```
#include <iostream>
using namespace std;
int main() {
    int num1,num2,number;
    cout << "Name:M.Ibrahim_72404" << endl;
    cout << "Enter your first number:";
    cin >> num1;

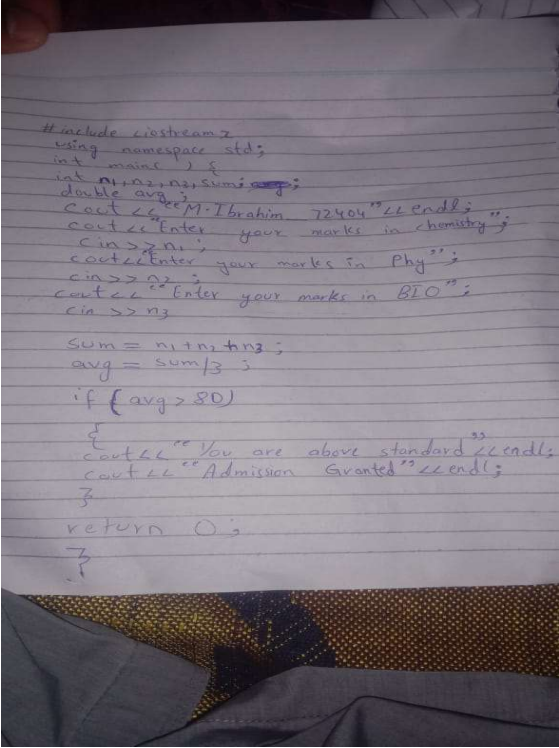
    cout << "Enter your second number:";
    cin >> num2;
    cout << "the number after swap are" << endl;
    number = num1;
    num1 = num2;
    num2 = number;
    cout << "The value of first number = " << num1
    << endl;
    cout << "The value of second number = " << number
    << endl;
    return 0;
}
```

•Q2

Write a program that input marks of three subjects, if the average of marks is more than 80 , it display two messages "You are above standard" and "Admission Granted".

```
using namespace std;
int main() {
    int n1,n2,n3,sum;
    double avg;
    cout<<"Name:M.Ibrahim_72404"<<endl;
    cout<<"Enter your marks in chem:";
    cin>>n1;
    cout<<"Enter your marks in phy:";
    cin>>n2;
    cout<<"Enter your marks in BIO:";
    cin>>n3;

    sum=n1+n2+n3;
    avg=sum/3.0;
    if(avg>80)
    { cout<<"You are above standard"<<endl;
      cout<<"Admission Granted"<<endl;
    }
```



```
#include <iostream>
using namespace std;
int main() {
    int n1,n2,n3,sum;
    double avg;
    cout<<"Name:M.Ibrahim_72404"<<endl;
    cout<<"Enter your marks in chemistry:";
    cin>>n1;
    cout<<"Enter your marks in Phy:";
    cin>>n2;
    cout<<"Enter your marks in BIO:";
    cin>>n3;

    sum = n1+n2+n3;
    avg = sum/3;
    if (avg>80)
    {
        cout<<"You are above standard"<<endl;
        cout<<"Admission Granted"<<endl;
    }
    return 0;
}
```

Output

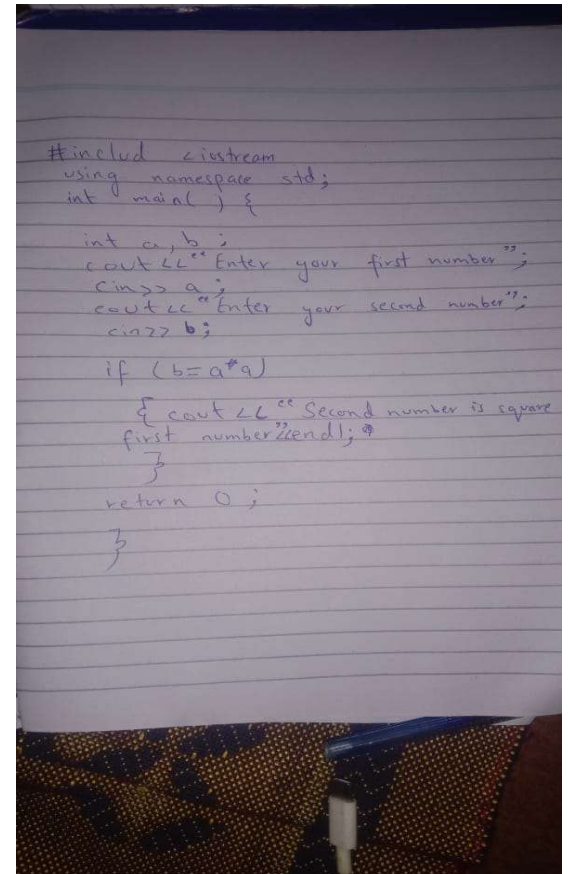
```
Name:M.Ibrahim_72404
Enter your marks in chem:89
Enter your marks in phy:90
Enter your marks in BIO:89
You are above standard
Admission Granted
```

Q3

Write a program that input two numbers and finds if second number is square of first number.

```
#include <iostream>
using namespace std;
int main() {
    int a,b;

    cout<<"Name:M.Ibrahim_72404"<<endl;
    cout<<"Enter your first number:";
    cin>>a;
    cout<<"Enter your second number:";
    cin>>b;
    if(b==a*a)
    {
        cout<<"second number is square of first number:"<<endl;
    }
    return 0;
}
```



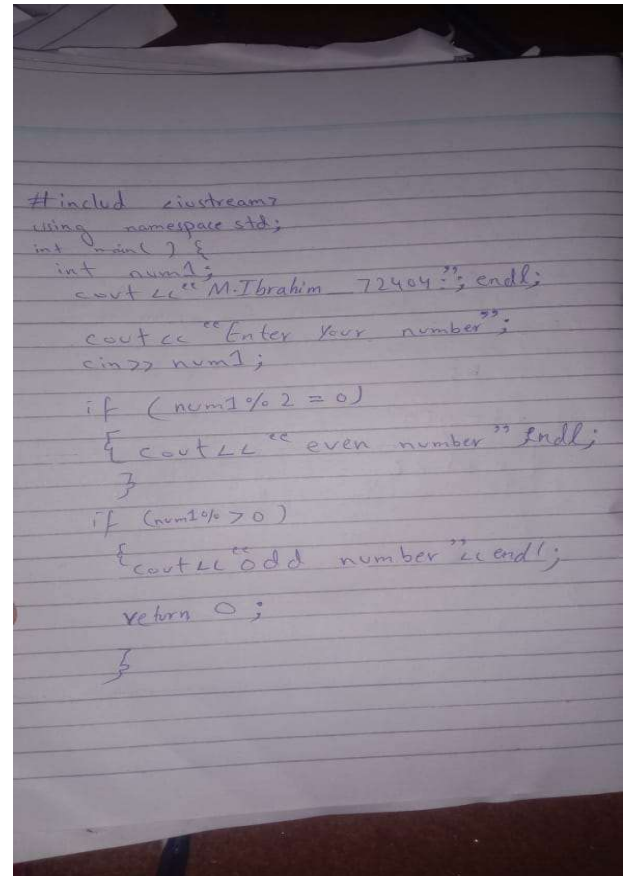
A photograph of a person's hand writing the same C++ code as shown in the code block on the left. The handwriting is in blue ink on lined paper. The code is:
#include <iostream>
using namespace std;
int main() {
 int a,b;
 cout<<"Enter your first number:";
 cin>>a;
 cout<<"Enter your second number:";
 cin>>b;
 if (b==a*a)
 {
 cout<<"Second number is square
first number"<<endl;
 }
 return 0;
}

```
Name:M.Ibrahim_72404
Enter your first number:2
Enter your second number:4
second number is square of first number:
```

Q 4 Write a program that input a number and find wheather the number is even or odd.

```
#include <iostream>
using namespace std;
int main() {
    int num1;
    cout<<"Name:M.Ibrahim_72404"<<endl;
    cout<<"Enter your number:";
    cin>>num1;
    if(num1%2==0)
    { cout<<"Number is even:"<<endl;
    }
    if(num1%2>0)
    { cout<<"Number is odd"<<endl;
    }

    return 0;
}
```



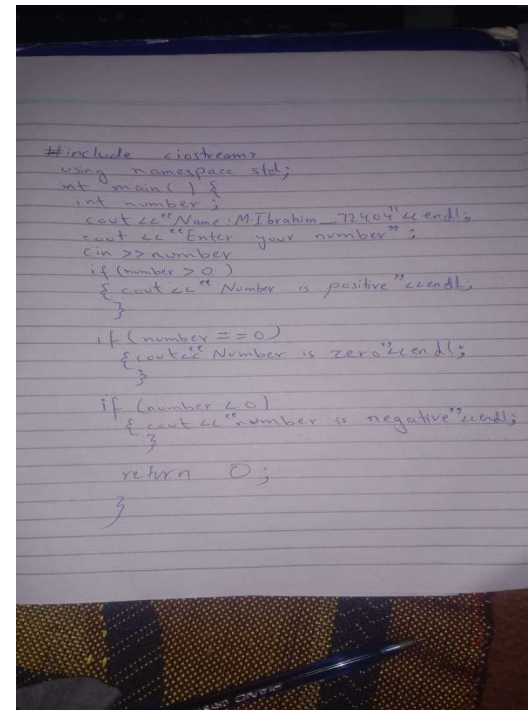
A photograph of a piece of lined paper with handwritten C++ code. The code is written in blue ink and matches the typed code in the first block. It includes the necessary headers, uses the std namespace, and implements a main function that prompts for a name and a number, then checks if the number is even or odd using modulo arithmetic. The code ends with a return statement and a closing brace.

```
#include <iostream>
using namespace std;
int main() {
    int num1;
    cout<<"M.Ibrahim_72404:"<<endl;
    cout<<"Enter your number:";
    cin>>num1;
    if (num1%2 == 0)
    { cout<<"even number"<<endl;
    }
    if (num1%2 > 0)
    { cout<<"odd number"<<endl;
    }
    return 0;
}
```

```
Name:M.Ibrahim_72404
Enter your number:8
Number is even:
```

Q 5: Write a program that input a number and determine it is positive , negative or 0.

```
#include <iostream>
using namespace std;
int main() {
    int number;
    cout<<"Name:M.Ibrahim_72404"<<endl;
    cout<<"Enter your number:";
    cin>>number;
    if(number>0)
    { cout<<"Number is positive:"<<endl;
    }
    if(number==0)
    { cout<<"Number is zero"<<endl;
    }
    if(number<0)
    {cout<<"Number is negative"<<endl;
    }
    return 0;
}
```



```
#include <iostream>
using namespace std;
int main() {
    int number;
    cout<<"Name:M.Ibrahim_72404"<<endl;
    cout<<"Enter your number:";
    cin>>number;
    if(number>0)
    { cout<<"Number is positive"<<endl;
    }
    if(number==0)
    { cout<<"Number is zero"<<endl;
    }
    if(number<0)
    { cout<<"Number is negative"<<endl;
    }
    return 0;
}
```

```
Name:M.Ibrahim_72404
Enter your number:-1
Number is negative
```

```
Name:M.Ibrahim_72404
Enter your number:5
Number is positive:
```

```
Name:M.Ibrahim_72404
Enter your number:0
Number is zero
```

•Q 6 Write a program that input three numbers and display maximum number

```
#include <iostream>
using namespace std;
int main() {
    int n1,n2,n3;
    cout<<"Name:M.Ibrahim_72404"<<endl;
    cout<<"Emter your first number:";
    cin>>n1;
    cout<<"Emter your second number:";
    cin >>n2;
    cout<<"Emter your third number:";
    cin>>n3;
    if(n1>=n2&& n1>=n3)
    {cout<<"maximum number is"<<n1<<endl;}
    if(n2>=n1&& n2>=n3)
    {cout<<"maximum number is"<<n2<<endl;}
    if(n3>=n1&& n3>=n2)
    {cout<<"maximum number is : "<<n3<<endl;
    }
}
```

```
Name:M.Ibrahim_72404
Emter your first number:4
Emter your second number:8
Emter your third number:7
maximum number is8
```